

Assignment 3

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COURSE: Cryptography and Network Security.

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HMAC with SHA-256

a) Purpose of HMAC

HMAC combines a secret key with a hash function to provide message integrity and authentication. Plain hash functions do not prove that the message came from an authorized sender.

b) HMAC Calculation Steps

Given:

Message: HELLO

Key: SECRET

Hash: SHA-256

Steps:

1. Convert the key and message into bytes.
2. Pad the key to the SHA-256 block size (64 bytes).
3. Create:

$\text{ipad} = 0 \times 36$ repeated 64 times
 $\text{opad} = 0 \times 5C$ repeated 64 times

4. Calculate the inner hash:

$$H(K \oplus \text{ipad} \parallel \text{HELLO})$$

5. Calculate the final HMAC

$$H((K \oplus \text{opad}) || \text{Inner Hash})$$

6. The resulting 256-bit value is the HMAC.

Formula:

$$\text{HMAC} = H((K \oplus \text{opad}) || H((K \oplus \text{ipad}) || M))$$

c) Why not MD5?

MD5 has known collision vulnerabilities and is considered cryptographically broken. Therefore, it should not be used for security-critical applications. SHA-256 is preferred.

d) MAC vs Digital Signature

MAC

Uses a shared secret key.

Provides authentication and integrity

Both parties know the same secret

Does not provide non-repudiation

Digital Signature

Uses private/public key pair

Provides authentication, integrity and non-repudiation.

Only sender has the private key

Provides non-repudiation.

HMAC Working in Detail

HMAC uses two important inputs:

1. Secret Key (K)
2. Message (M)

The general formula is:

$$\text{HMAC}(K, M) = H[(K \oplus \text{opad}) \parallel H((K \oplus \text{ipad}) \parallel M)]$$

Here:

H = cryptographic hash function

K = secret key

M = message

ipad = inner padding

opad = outer padding

$\oplus$ = XOR operator

$\parallel$ = concatenation

Advantages of HMAC

1. It provides message authentication
2. Provides data integrity
3. Uses a secret key, preventing unauthorized users from generating a valid MAC.

4. Can be used with different secure hash functions.

5. It is efficient and suitable for network communication.

6. It protects messages from unauthorized modification.

For Example

HELLO $\rightarrow$ SHA-256 $\rightarrow$ Hash value

With HMAC

HELLO + SECRET KEY $\rightarrow$ HMAC $\rightarrow$
Authentication Code

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